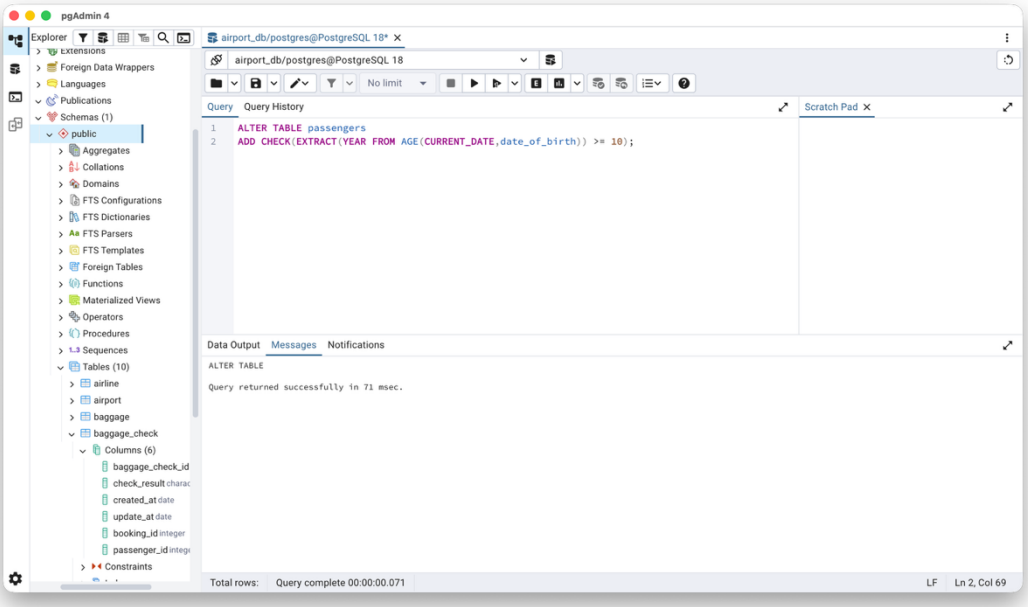
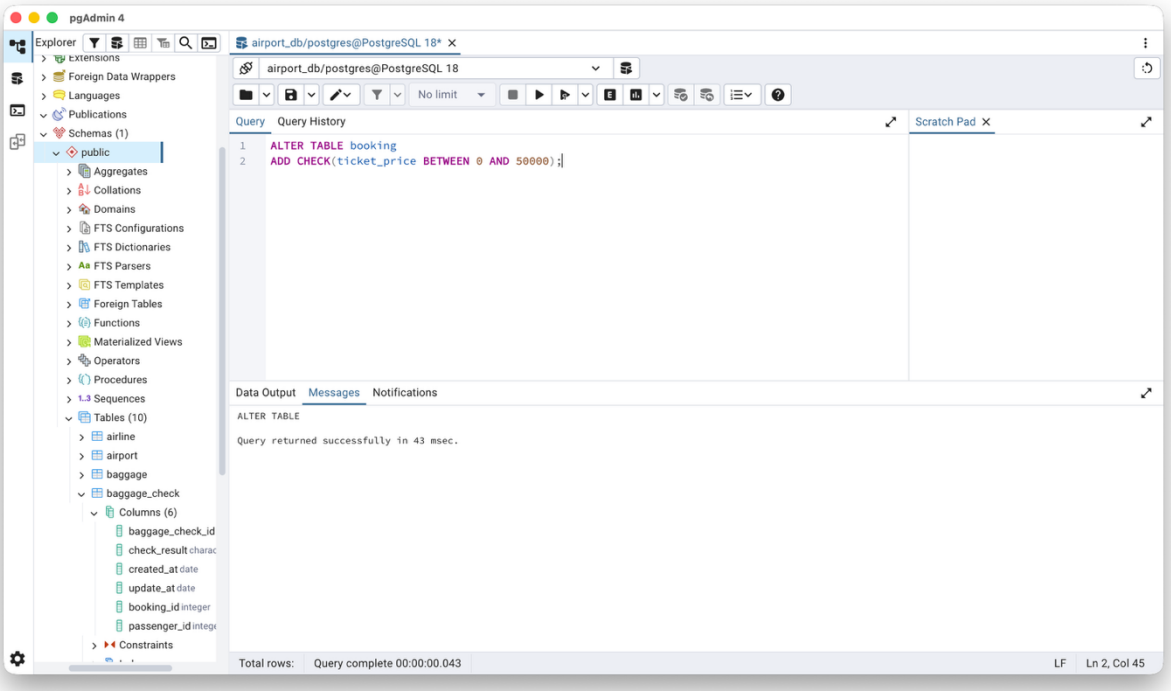
Laboratory work 5

Tasks:

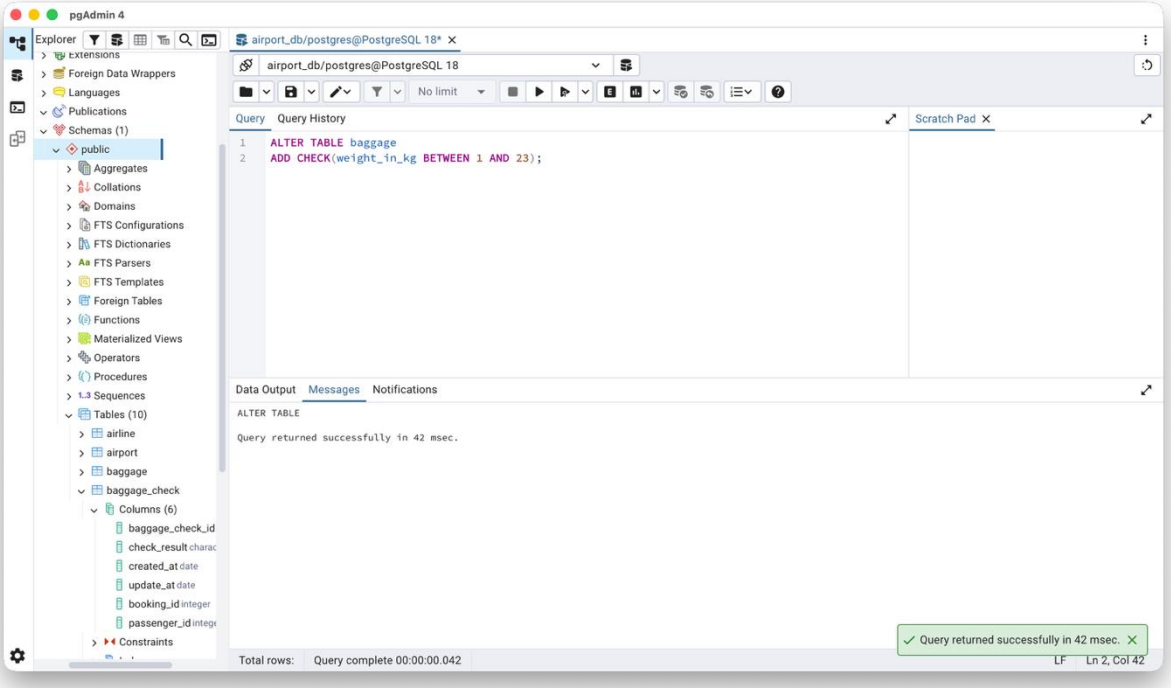
1. Add a CHECK constraint to passenger table to provide that passengers must be at least 10 years old.



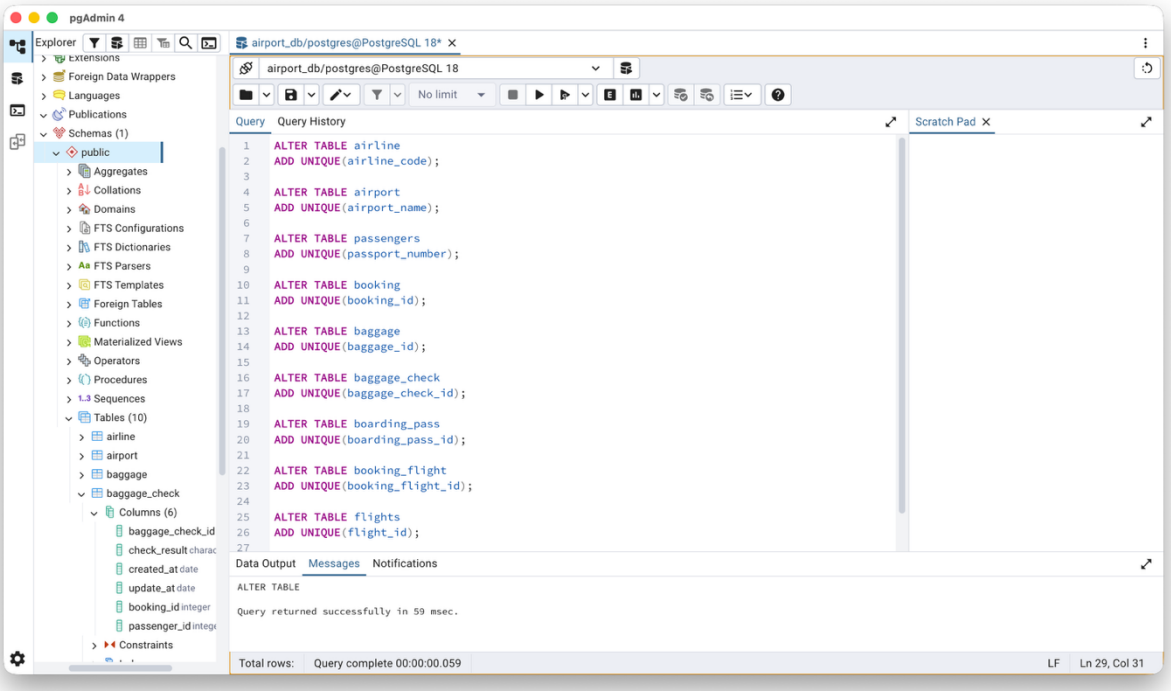
2. Add a CHECK constraint to accept values in booking price not more than 50000tg and less than 0tg.



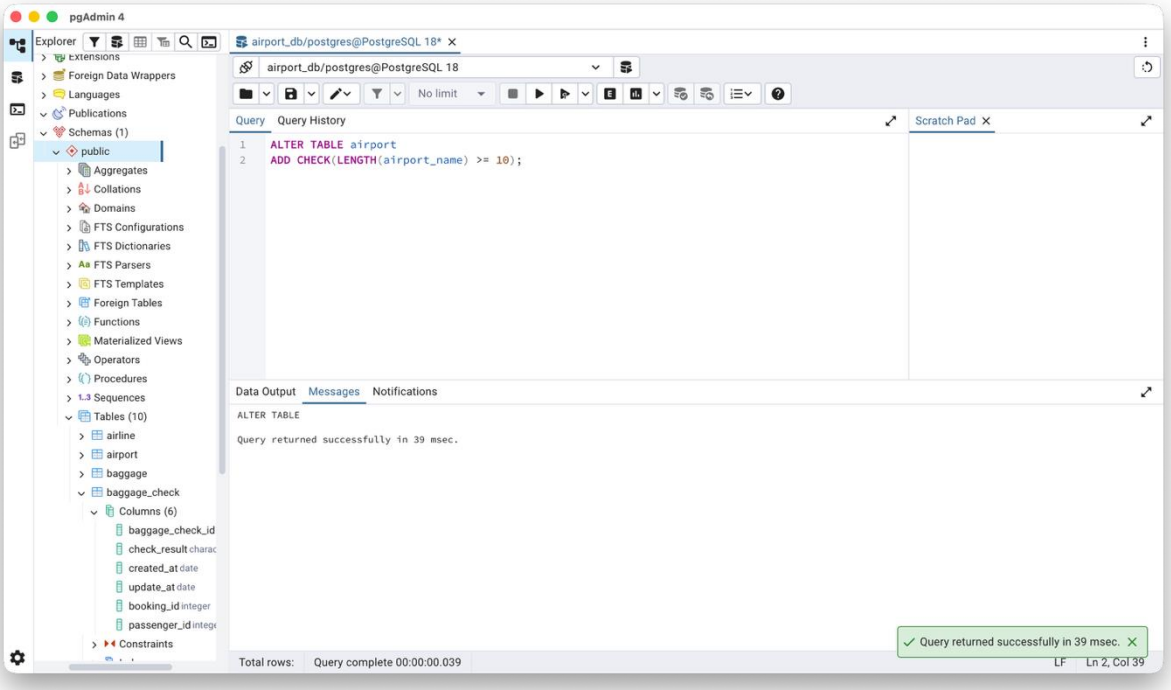
3. Add a CHECK constraint to accept the luggage weight between 1 and 23 kg.



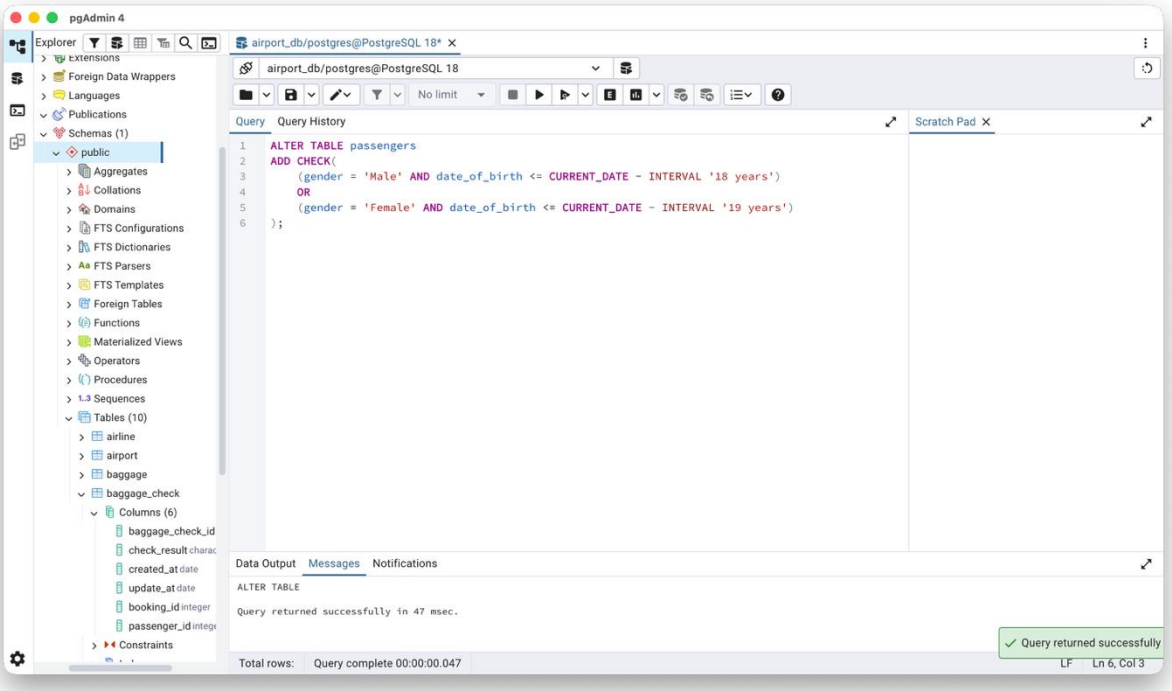
4. Add a CHECK constraint to ensure that all values in airport_name must have at least 10 characters.



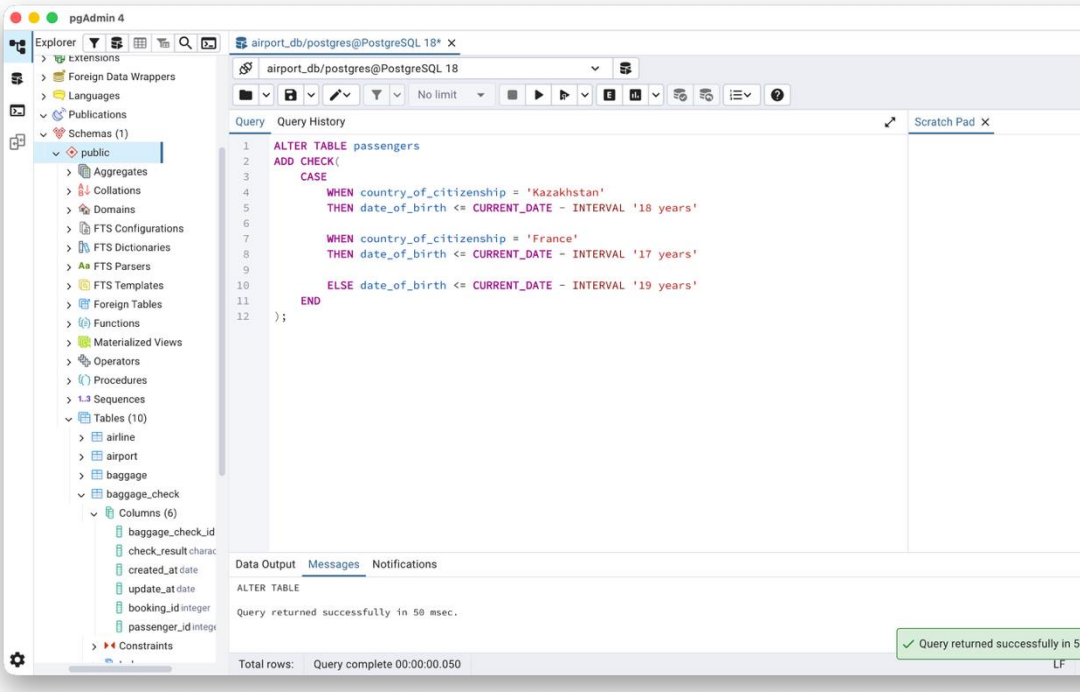
5. Add UNIQUE constraint to some columns in each table in database.



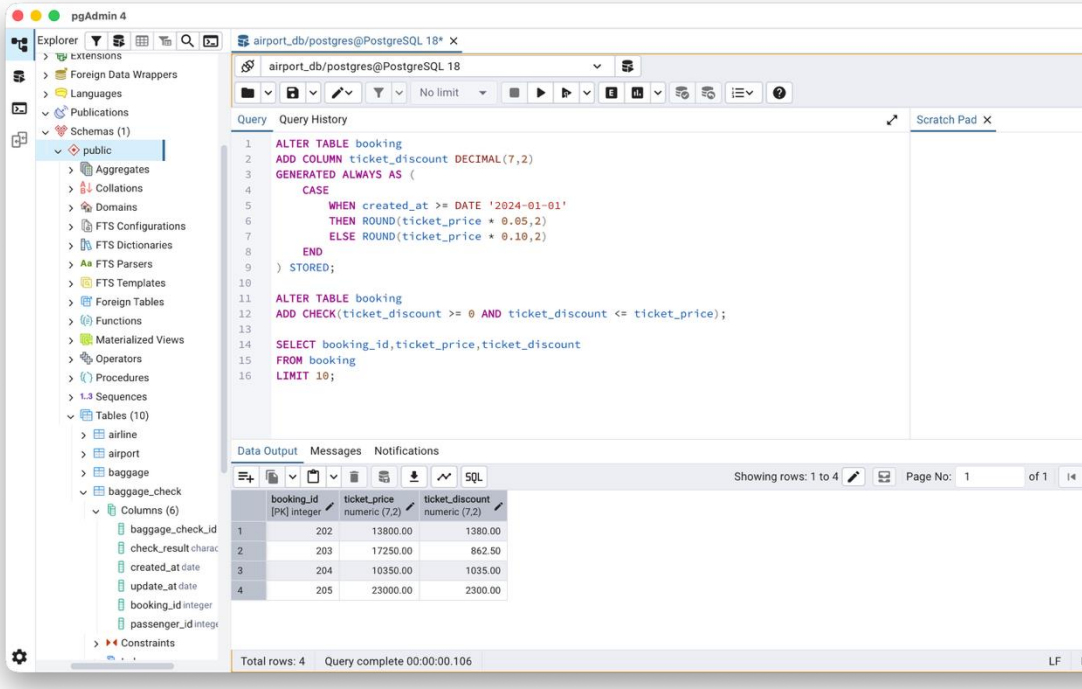
6. Add a CHECK constraint to ensure that male passengers must be at least 18 years old and female passengers must be 19 years old.



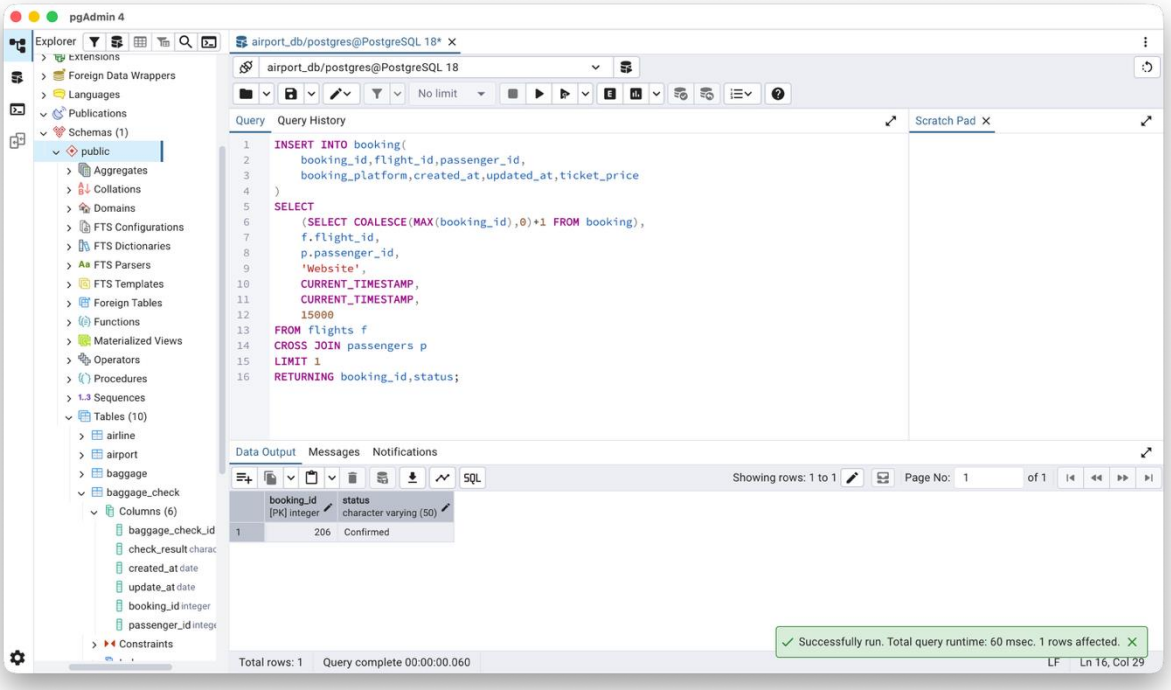
7. Add a CHECK constraint to add rule as follow (use column country_of_citizenship):
   - Passengers from Kazakhstan must be at least 18 years old.
   - Passengers from France must be at least 17 years old.
   - Passengers from other countries must be at least 19 years old.

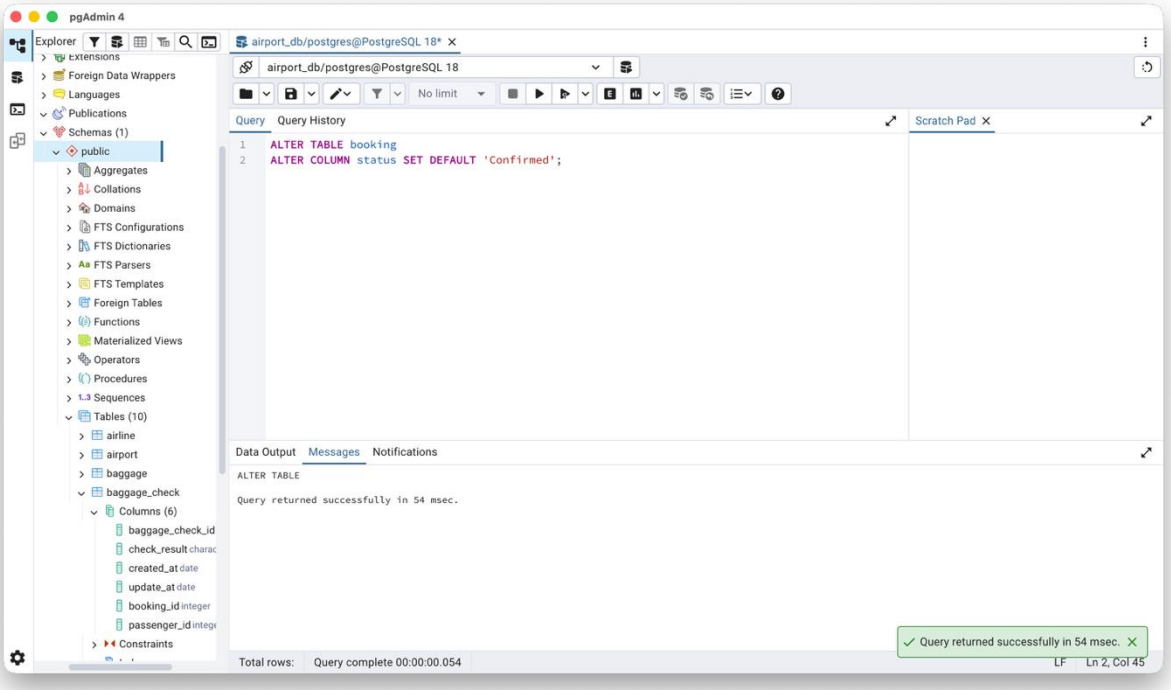


8. Add a ticket_discount column to table booking and a CHECK constraint to apply some discount based on ticket price and created time:
    - the constraint applies a 5% discount for tickets created after 2024-01-01,
      and 10% discount for tickets created before 2024-01-01.



9. Modify the booking table so that the booking status column has a default value of 'Confirmed'. Then insert a new booking without specifying booking status.


pgAdmin 4
Explorer
airport_db/postgres@PostgreSQL 18*
airport_db/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
INSERT INTO booking(
booking_id,flight_id,passenger_id,
booking_platform,created_at,updated_at,ticket_price
)
SELECT
(SELECT COALESCE(MAX(booking_id),0)+1 FROM booking),
f.flight_id,
p.passenger_id,
'Website',
CURRENT_TIMESTAMP,
CURRENT_TIMESTAMP,
15000
FROM flights f
CROSS JOIN passengers p
LIMIT 1
RETURNING booking_id,status;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 1
Page No: 1
of 1
booking_id
[PK] integer
status
character varying (50)
1
206
Confirmed
Successfully run. Total query runtime: 60 msec. 1 rows affected.
Total rows: 1
Query complete 00:00:00.060
LF
Ln 16, Col 29


pgAdmin 4
Explorer
airport_db/postgres@PostgreSQL 18*
airport_db/postgres@PostgreSQL 18
No limit
Query
Query History
Scratch Pad
ALTER TABLE booking
ALTER COLUMN status SET DEFAULT 'Confirmed';
Data Output
Messages
Notifications
ALTER TABLE
Query returned successfully in 54 msec.
Query returned successfully in 54 msec.
Total rows:
Query complete 00:00:00.054
LF
Ln 2, Col 45

10. Modify the appropriate date column in the database so that when a new record is inserted, the date is automatically set to the current date.

